QUESTION 22

Domain-Specific Chatbot Using LangChain and a Vector Database

Domain: Artificial Intelligence and Machine Learning

Framework: LangChain | Vector Database: FAISS | LLM: FLAN-T5

1. Aim

To develop an interactive domain-specific chatbot using LangChain and a vector database that retrieves relevant domain knowledge and generates answers to user questions.

2. Problem Statement

General-purpose chatbots may provide broad responses that are not restricted to a selected knowledge domain. A domain-specific chatbot improves relevance by searching a controlled knowledge base before generating its response.

3. Objectives

- Create an Artificial Intelligence and Machine Learning knowledge base.
- Represent domain information using LangChain Document objects.
- Generate semantic embeddings for domain documents.
- Store document embeddings in FAISS.
- Use a LangChain retriever to identify relevant topics.
- Generate responses using an LLM.
- Support multiple questions during one chatbot session.

4. System Architecture

Domain Knowledge → LangChain Documents → Embeddings → FAISS Vector Store → LangChain Retriever → User Question → Retrieved Context → LLM → Chatbot Response

5. Technology Stack

Component	Implementation
Language	Python
Framework	LangChain
Embedding Model	all-MiniLM-L6-v2
Vector Database	FAISS
Retriever Top-K	3
Generation Model	google/flan-t5-small
Domain	Artificial Intelligence and Machine Learning

6. Working Procedure

1. Load domain knowledge as LangChain Document objects.

2. Create the embedding model.

3. Build the FAISS vector store from domain documents.

4. Create a LangChain retriever configured for Top-3 retrieval.

5. Accept a question from the user.

6. Retrieve the most relevant domain topics.

7. Build context from retrieved documents.

8. Generate a chatbot response using FLAN-T5.

9. Display retrieved topics and the generated response.

10. Continue the conversation until the user exits.

7. Multiple-Question Chatbot Demonstration

The implementation was tested with three different questions in the same interactive session. This demonstrates that the chatbot can process successive questions without restarting the application.

Question	Ranked Domain Topics	Chatbot Response	Observation
What is machine learning?	1. Machine Learning 2. Supervised Learning 3. Deep Learning	a branch of artificial intelligence that enables computers to learn patterns from data	Correctly retrieved the Machine Learning topic and generated a relevant definition.
What is supervised learning?	1. Supervised Learning 2. Machine Learning 3. Unsupervised Learning	a branch of artificial intelligence that enables computers to learn patterns from data	Supervised Learning was ranked first, but the generated answer was broader than the retrieved topic.
What is NLP?	1. NLP 2. Supervised Learning 3. Reinforcement Learning	Natural language processing	NLP was ranked first and the chatbot correctly expanded the abbreviation.

8. Actual Chatbot Session

You: What is machine learning?

Retrieved Domain Topics:
1. Machine Learning
2. Supervised Learning
3. Deep Learning

Chatbot:
a branch of artificial intelligence that enables computers to learn patterns from data

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You: What is supervised learning?

Retrieved Domain Topics:
1. Supervised Learning
2. Machine Learning
3. Unsupervised Learning

Chatbot:
a branch of artificial intelligence that enables computers to learn patterns from data

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You: What is NLP?

Retrieved Domain Topics:
1. NLP
2. Supervised Learning
3. Reinforcement Learning

Chatbot:
Natural language processing

9. Retrieval Analysis

For all three questions, the most relevant domain topic was ranked first: Machine Learning for the first question, Supervised Learning for the second, and NLP for the third. This confirms that the LangChain retriever is selecting semantically related domain knowledge.

10. Response Analysis

The NLP query produced the most precise response, "Natural language processing." The machine learning query also produced a relevant definition. For the supervised learning query, the retriever correctly identified Supervised Learning as the first topic, although the generated answer repeated a broader machine-learning definition. This shows that retrieval and generation are separate stages: retrieval can be correct even when the small generation model produces a less specific answer.

11. Result

The domain-specific chatbot successfully processed three consecutive questions in one session. It retrieved Top-3 domain topics for each question and generated responses using the retrieved knowledge and FLAN-T5. The results demonstrate semantic domain retrieval and interactive chatbot operation.

12. Advantages

- Domain-focused knowledge retrieval.
- Semantic rather than exact keyword search.
- Interactive multi-question operation.
- Easy to extend by adding new domain documents.
- Clear separation between retrieval and generation.

13. Applications

- Educational assistants

- Technical support chatbots
- University knowledge systems
- Research assistants
- Enterprise knowledge assistants
- Domain-specific help systems

14. Conclusion

A domain-specific chatbot was successfully implemented using LangChain, FAISS, semantic embeddings, and a language model. The chatbot handled three consecutive questions about machine learning, supervised learning, and NLP. The retriever consistently ranked the most relevant domain topic first, demonstrating effective semantic retrieval. The implementation provides a practical foundation for focused AI assistants.